



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Writing Successful FYP Reports

Prof Lucy Hederman

Outline

This week:

- I. **Preparing the ground**
- II. **Planning the report**

Next time:

- III. **Writing**

I. Preparing the ground

a. **Expectations**

- What is expected of your FYP report?

b. **Information Management**

- Keep in mind the goal as you gather info and data

c. **Time and Space Management**

d. **Practice Writing**

- Write early and often

I.a Expectations

- General expectations of academic writing
 - Coherent development of thesis
 - Appropriate structure and signalling
 - Evidence of coverage and understanding of literature
 - Rigour, sound reasoning in analysis, discussion
 - Appropriate methodology properly executed
 - Competent presentation - writing, graphs, etc.
- Specific guidelines for your degree
 - <https://projects.scss.tcd.ie/information-for-students/report-and-dissertation-guidelines/>
- Previous FYP reports for size, formatting, structure, style

FYP criteria for “Exceptional (80%+)”

Problem statement, motivation, and analysis

“Demonstrates a thorough understanding of the problem area. The work is clearly motivated, with deep critical analysis of the problem. The project objectives should be clearly stated and given an insightful and critical analysis.”

Background Research & Literature Review

“Excellent review of the literature demonstrating extensive knowledge of past/current work/authors in the field. Critical and analytical perspective, data collection & analysis. For an undergraduate the report demonstrates an exceptional ability to relate theoretical knowledge to project work.”

FYP criteria for Exceptional ...continued

Technical content and project execution

Comprehensive development of the technical theme, including an element of originality. Appropriate choice of strategy and approach. Limitations recognised. An ambitious project executed well.

Testing, evaluation, critical analysis & conclusions

Work of exceptional quality. Exemplary presentation and analysis of results, evidence of sound critical evaluation. Limitations/future research areas clearly defined.

Report presentation and writing

Report is excellently written, coherent, internally consistent, well organized and excellently presented. The report adheres throughout to academic conventions with respect to formatting and referencing and is carefully and effectively presented. Excellent use of figures, tables, and diagrams where appropriate.

Previous FYP reports

<https://www.scss.tcd.ie/publications/projects/>

For internship students:

No previous reports shared. But the marking scheme can be shared.

Tang, Neasa (2018/19) [PDF](#) [ABSTRACT](#)

BA (Mod) Computer Science

"AVERT System Prototype"

(Supervised by: Hederman, Lucy)

Whelan, Brian (2018/19) [PDF](#) [ABSTRACT](#)

BA (Mod) Computer Science

"An Investigation into Locomotion and Interaction Mechanics in a Environment"

(Supervised by: McDonnell, Rachel)

Whittam, Joanna (2018/19) [PDF](#) [ABSTRACT](#)

BA (Mod) Computer Science

"Visual Measurement of Chest Compression Depth in Cardiopulm"

(Supervised by: Dawson-Howe, Kenneth)

BA (Mod) Computer Science and Business (2018/19)

Allan, Paul (2018/19) [PDF](#) [ABSTRACT](#)

BA (Mod) Computer Science and Business

"Analysing Social Media Posts and Interactions through Sentiment Language Processing"

(Supervised by: Arnedillo, Inmaculada)

Cox, James (2018/19) [PDF](#) [ABSTRACT](#)

BA (Mod) Computer Science and Business

"Data Protection Process Browser Widget: A Visualization Tool for Privacy Policies"

(Supervised by: Lewis, Dave)

I.b Information Management

Prior to writing you

- Gather facts and figures
- Read related ideas
- Study relevant methodologies
- Examine how others write
- Measure or design and implement and evaluate (primary research)

All these activities should be done in such a way as to facilitate productive writing.

When you read

Have reading goals

- Why am I reading this? What do I hope to get from it
 - Data
 - Ideas related to my topic
 - Ideas for how to structure a discussion
 - Examples of methodologies

Read critically

- Is this author convincing? Why? Why not?
- Make frequent margin notes; follow hunches.

When data gathering

Define goals for the task at hand

- Before interviewing, surveying,

Write-up your goal and data-gathering plan

- Convince yourself it's coherent, rigorous.
- Perhaps run it by supervisor or colleague.

Maintain careful complete records

- of data or background info sources
- of experiment conditions

During design and implementation

Document your plan and its evolution

- Some summary of this will make it into your report

Keep notes of design decisions

- All too easy to forget why you did it this way

Record key implementation challenges

- All too easy to forget the interesting stuff when it comes time to write up

Use a (physical) notebook for all your FYP ideas and thoughts and planning.

I.c Space and Time

Dedicated space if possible.

Regular planned sessions if possible

- E.g 7-9am two mornings and 10-12 Sunday might work for you.
- At end of one session knowing when the next one will be seems to ease anxiety.
- At end of each session list actions for next session (in your notebook).
 - Unconscious may progress these for you

More on scheduling later.

I.d Practise Writing

Write early and often

- Aids clear thinking
- Develops fluency and ease
- Allows more useful feedback
- Feeds into final draft

Aim to write something in advance of every supervisor meeting

- Proposal, plan of next steps, overview of literature, preliminary design, ...
- Send it in advance
- Solicit specific feedback in advance
 - “Does the discussion on p.3 adequately capture the ...?”

Summary of I

**Preparation for
the writing
starts now.**

Writing Task 1

Take 5 minutes to write fluently on a project you completed previously

- Mention objective, goals, design, implementation, evaluation.
- For private consumption.
- Don't worry about structure or style now.

II. Planning

- a. **Whole text development**
- b. **Sequential outlining**
- c. **Planning document production**

II.a Whole Text

What do I want to say?

- Main message
- Subsidiary messages

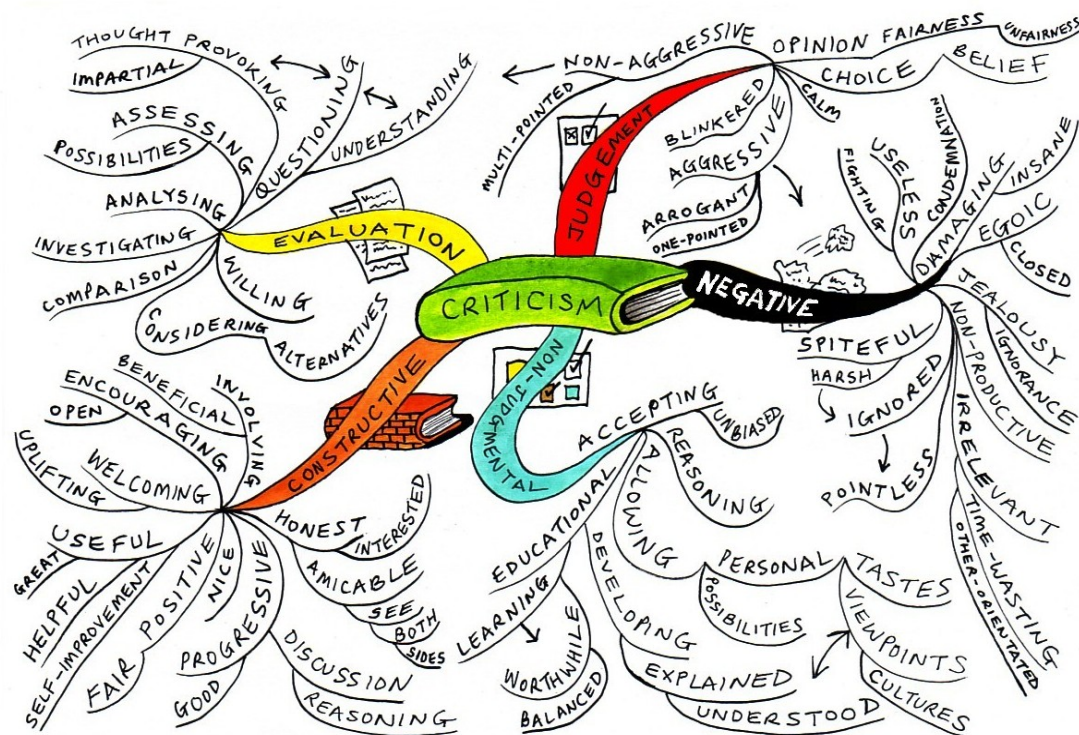
How do I tell the story?

- Coherence
- Ordering

What do I want to say?

Brainstorming/mindmapping

- Can be used for whole text and for each chapter and main sections
- Start with research question, or a chapter title
- No ordering
- Ignore development of argument issues



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Task 2: Mindmapping

Stop the video here and spend five minutes:

Taking the focus of your early piece of writing as the central topic, develop a mind map of that topic with a view to generating a possibly better version of your piece next time.

How to say it?

How to order your messages for best effect

- Appropriateness of information for particular parts
- Logical relations between parts
- Reader's need for information

How to link messages to expose the coherence

- “The previous chapter concluded that This chapter addresses ...”
- “Given the problems outlined in the previous section, some solutions that have been adopted are ...”
- Sentence-level: But, however, therefore,

What, Why, How, So what?

These questions can help at the brainstorming stage, and also in writing coherent, convincing text.

They can be applied at many levels

- “How” can be interpreted in a number of ways; think about which ones are helpful in your context.

At dissertation level:

- What is my report about?
- Why did I do it?
- How did I carry it out?
- So what – what’s the contribution, value?

What, Why, How, So what

At the chapter level

- What is this chapter about?
- Why is this chapter here?
- How was the material gathered and how is it organised and presented?
- So what? What is the message to be taken from this chapter?

What, Why, How, So what

In a section of a literature review

- **What** does this section cover?
- **Why** am I covering it, what's the relevance to my research?
- **How** is the material handled?
 - e.g. “first the origins of X are presented, then a number of parallel developments are compared and finally current best practice is summarised.”
- **So what** does this mean for my dissertation?

Task 3: Coherence

Stop the video now to:

Identify the main ideas in your project description text.

- Circle them or note the key topics in the margin

Underline linking texts, phrases

Could you order your main ideas better?

Could you link your ideas better?

Summary of II.a

**Think about your
messages
and how best to
get them across
to your reader**

II.b Sequential Outlining

Once you've identified all the main topics

Plan the ordering and flow

- For coherent argument
- So necessary material is explained before needed

Include chapter titles, objective of the chapter, main sections, word count

Typical Research Report Outline

Preliminary pages

Chapters – normally 5 to 7

- Chapter 1 ~ The Introduction
- Chapter 2 ~ Literature Review
- Chapter 3 ~ Methodology
- Chapter 4 ~ The Research
- Chapter 5 ~ The Findings
- Chapter 6 ~ Conclusions, Limitations and Future Research

References

(Appendices)

Common CS style Outline

Preliminary Pages

Chapters

- Introduction
- (Background)
- State of the Art
- Design
- Implementation
- Evaluation & Discussion
- Conclusions

References

(Appendices)

Chapter 1: Introduction

Brief context for problem

Motivation

Research Question / Project objectives

Outline of the project work

Outline of the report

The intro sets up the report but doesn't include any real meat.

Aim for 4-5 pages max.

Chapter: Background

In some cases there is substantial Background/Context material that is not really “State of the Art”.

- State of Art should be mostly about other efforts to solve your problem or related ones.

In such cases add a chapter here

- It might have a specific title like “Linked Data”
- It often extends the Intro

Chapter: State of the Art

Aka Literature Review

Critical review

- Analysis and synthesis

Relevant to research question

- Not just a brain dump!

Structured, with summaries/conclusions.

- Avoid writing about one piece of research after the other. Find the connections and work them in.

More guidance next time.

Chapter: Methodology

Structure of the study

What approach you took to answering the research question

You should also explain why you took this approach and why it is better than other approaches you know about and decided not to use.

For a software project this might describe the requirements and design of the software.

Chapter: Design

(For Software projects)

How you went from problem statement (in the intro) to the design of your thing.

- **Synthesis of existing solutions**
- **Requirements gathering**
- **Design phases**
- **...**

Chapter: The Research

What was actually done and how it was done

This chapter describes the carrying out of the

- Questionnaire
- Interviews
- Experiment/s
- Focus groups
- Action research
- Software implementation

What was the outcome of the research i.e. what data or evidence did you acquire?

Chapter: Implementation

What was actually done and how it was done

- Architecture (if this was not a key part of your design)
- Tools and platforms used (and why)
- How key implementation challenges were addressed

Chapter: Evaluation & Discussion

What did you do to test, validate and evaluate your work?

How well did the project meet the objectives?

What analysis did you do to help you understand the data or evidence?

What are the limitations of your work?

Chapter: Conclusion

Brief review

Main results

- Related back to project objectives
- Summary of any proposals
- Assessment of contribution (So what?)

Future work

- What you would do if you had more time
- Where someone else could go with this work

Reflection

- What did you learn; what went well, wrong ...
- what do you think about your research and findings and contribution

References

Always acknowledge the sources of all your quotations, arguments and ideas, even if you don't quote the source directly.

Complete

- Reader must be able to find the source
- Include all authors – use “et al.” in citation labels, but not in the list of references..
- Where an article has been published on the Web and in print, use the print reference in preference

You are not stuck for space, so

- Use meaningful tags, not just numbers (e.g. Harvard style)
- <http://www.tcd.ie/library/support/subjects/nursing-midwifery/assets/Harvard%20Ref%20System%202016-2017.pdf>
- Don't abbreviate Journal titles, etc.

Appendices

For supplementary material which is not necessary in order for the reader to follow the argument.

- the text of a questionnaire, full results, detailed UML diagrams, or a complete Software Requirement Specification

Structuring Material in Chapters

Before writing, plan each chapter as a series of sections.

- What's the main message of each one?
- What sub-sections are required?

Each section should have

- an introduction that tells us what its role is in the whole
- a summary or conclusion that pulls together the main message(s) of the section.

Number sections and subsections hierarchically

- 3.1, 3.2.4, etc.

Summary of II.b

**Plan your
report outline
(chapters and main sections)
early**

II.c Planning Doc Production

Managing drafts

Document production software

Managing references

Managing drafts

Use page numbers, file name and draft date on all documents and drafts.

Have a disaster recovery plan!!!

- Don't depend on your hard drive and a memory stick!
- Use Dropbox or similar but be careful ...
 - Automatic syncing is great till you save an empty doc over your only version ...

Plan document production

MS Word or Latex?

Latex

- **Overleaf**

If using MS Word

- Consider going on a Word course.
- **Use formatting Styles to control format.**

Managing References

Choose a reference manager that will automatically insert references into your document and generate a bibliography.

Consider EndNote, Mendeley, Zotero or similar

- Comparison at <http://blogs.plos.org/mfenner/files/2010/09/Reference-Manager-Overview.png>
- Library training for EndNote X6
<http://www.tcd.ie/Library/support/endnote/training.php> or use EndNote Web.

Start using it!!

- Ensure full details of every relevant source is recorded in your reference manager when you find it, not when you decide to cite it.

Summary on IT Planning

There is lots you can do before you write a word!

- Identify the main messages
- Dissertation structure
- Document production software
- Managing references

Reflect

Before you go anywhere, write down (in your FYP notebook) the key messages you want to remember from this week's video sessions.

Sources

Gail Craswell, *Writing for Academic Success*, Sage Study Skills, 2005

- lots of very specific advice

Murray, R. and S. Moore, *The Handbook of Academic Writing, A Fresh Approach*, Open University Press, 2006

- developing the writing habit

www.uefap.com

- Using English For Academic Purposes

Some online tools

Hemingway for writing style

Grammarly.com for advanced grammar checking

writefullapp.com for checking how often your phrase or text appears on Google, and so is it correct?, and suggest words for contexts, etc.?

Scrivener for full toolkit for a dissertation or other long document.

See others on <http://www.producthunt.com/e/tools-for-writers>



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Thank You